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Source: *Physiological Zoology*, Jul. - Aug., 1990, Vol. 63, No. 4 (Jul. - Aug., 1990), pp. 757-776

Published by: The University of Chicago Press. Sponsored by the Division of Comparative Physiology and Biochemistry, Society for Integrative and Comparative Biology

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The Metabolic Cost of Maintaining Position for Four North American Stream Fishes: Effects of Season and Velocity

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Accepted 10/26/89

Abstract

We used a swimming respirometer to measure oxygen consumption (i.e., metabolic cost) of four species of stream fishes while they held position at different velocities. We tested naturally acclimatized individuals during spring, summer, fall, and winter at seasonal temperatures and photoperiods. Rosyside dace (Clinostomus funduloides), longnose dace (Rhinichthys cataractae), mottled sculpin (Cottus bairdi), and juvenile rainbow trout (Oncorhynchus mykiss) all showed seasonal changes in oxygen consumption. Seasonal changes in standard metabolic rates seemed to be influenced by changes in reproductive condition as well as ambient temperature. The metabolic cost of maintaining position for rainbow trout and rosyside dace generally increased with velocity. At velocities below eight body lengths per second, longnose dace did not exhibit a significant change in oxygen consumption, because they held position without swimming. At higher velocities, however, longnose dace swam to maintain their position and oxygen consumption increased rapidly. Mottled sculpin held position without swimming and therefore showed little or no change in energy use with changing current velocity.

Introduction

The energetic requirements of fishes living in streams can be considerable. The metabolic costs incurred by stream fishes can be estimated by measuring their rates of oxygen consumption at different velocities (providing anaerobiosis is not a significant contributor to overall metabolism) and then applying an appropriate oxycaloric equivalent. Such research has greatly

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enhanced our understanding of the behavior and physiological ecology of stream fishes. For example, Feldmeth and Jenkins (1973) used the respirometry data of Rao (1968) to estimate the energy expenditure of rainbow trout (*Oncorhynchus mykiss*) in a mountain stream, and Fausch (1984) applied equations developed by Stewart (1980) to estimate the profitability of stream microhabitats for coho salmon (*Oncorhynchus kisutch*) and rainbow trout. In addition, Alley (1977) and Smith and Li (1983) linked energetic demands to microhabitat use in several stream fishes.

Seasonal physiological changes may affect such estimates of metabolic costs. Metabolic rates of fishes have been shown to increase as acclimation temperature increases (see reviews by Brett [1971] and Fry [1971] and more recent studies by Caulton [1978]; Kruger and Brocksen [1978]; Ott, Heister, and Ultsch [1980]; Duthie [1982]; Vondracek, Cech, and Longanecker [1982]; Cech et al. [1985]; Eccles [1985]; Lemons and Crawshaw [1985]; DuPreez, Strydom, and Winter [1986]; and McClanahan et al. [1986]). Seasonally varying factors besides temperature, however, are also capable of producing changes in oxygen consumption (Wells 1935; Roberts 1964; Dickson and Kramer 1971; Fry 1971; Burns 1975; Evans 1984). Hence, energetic estimates based on oxygen consumption data from fishes acclimated to different temperatures during one season may not accurately represent seasonal effects (see Evans 1984).

We were interested in estimating the metabolic costs incurred by stream fishes while holding position in a current. Strategies for holding position differ among species. For example, water-column fishes hold position by swimming whereas benthic species may simply cling to the substrate. To evaluate the energetic costs associated with these strategies we measured oxygen consumption rates at different velocities for two water-column fishes, rainbow trout and rosyside dace (*Clinostomus funduloides*), and two benthic fishes, longnose dace (*Rhinichthys cataractae*) and mottled sculpin (*Cottus bairdi*). These species are abundant in southern Appalachian streams, and much is known about their patterns of microhabitat use (Grossman and Freeman 1987). We captured fish of each species during different seasons and held and tested them at seasonal temperatures and photoperiods. Our data, therefore, should accurately quantify seasonal costs of maintaining position. Although several investigators have quantified metabolic rates of rainbow trout (Rao 1968; Dickson and Kramer 1971; Ott et al. 1980; Wieser, Platzer, and Hinterleitner 1985), to our knowledge published metabolic data on the other three species are not available. Consequently, we attempted to answer the following questions in the study: (1) does oxygen consumption (i.e., metabolic cost) change as these fishes hold position against increasing water velocity, and (2) does the relationship between water velocity and oxygen consumption change seasonally?

Material and Methods

We obtained specimens for respirometry from Coweeta Creek, N.C., by direct-current electrofishing. For a detailed description of this system see Grossman and Freeman (1987). Fishes were captured each season and held at temperatures and photoperiods that approximated ambient conditions: spring (March 16–June 15; 10 °C, 12L:12D); summer (June 16–September 15; 15 °C, 14L:10D); fall (September 16–December 15; 10 °C, 12L:12D); winter (December 16–March 15; 5 °C, 10.5L:13.5D). Each daylight period began and ended with 0.5 h of low light to simulate twilight. We held fishes in 165-L flow-through tanks of dechlorinated tap water for at least 14 d before experiments (mean $\pm$ 1 SD = 27 d $\pm$ 13). Fishes readily accepted a diet of frozen brine shrimp and commercial trout chow and were fed to satiation daily. Specimens were not fed for 20 h prior to respirometry trials, but we did not examine individuals to determine whether they were in a postabsorptive state.

Estimates of Metabolic Costs

We used a modified Brett-style (Brett 1964) swimming respirometer (total volume 9.75 L) to measure oxygen consumption of fishes. Stainless steel screens confined fishes to a 20-cm-long swimming compartment with a 72-cm² cross-sectional area. Plexiglas grids upstream and downstream of the swimming compartment minimized water turbulence. A dye-movement study indicated that water circulated throughout the swimming compartment with minimal turbulence. We shaded the upstream half of the swimming compartment with a piece of dark cloth to provide visual orientation for test fish and to encourage them to remain in this portion of the compartment.

We conducted respirometry experiments at the same temperatures and photoperiods at which fishes were held. Individual fish were maintained in continuously circulating fresh water in the respirometer for at least 16 h before an experiment. Three hours before the first velocity trial, a current of approximately 1 cm s⁻¹ was initiated in the respirometer. Just before the first trial, we gradually increased the velocity to 5 cm s⁻¹ for 15 min. The water exchange pump was then turned off and the dissolved oxygen (DO) content of the water in the respirometer monitored for 30 min. We ran swimming trials at 10 cm s⁻¹ increments from 5 to 75 cm s⁻¹. Total oxygen content rarely fell below 95% of saturation and consequently should not have significantly affected metabolic rates or swimming performance (Davis et al. 1963; Kutty 1968). We tested at least 10 individuals of each species during each season.

We used a Yellow Springs Instruments (YSI model 58) oxygen meter to measure DO in the respirometer. The precision of the oxygen meter was checked twice during the course of the study. Dissolved oxygen readings were taken every minute during a trial. Between trials we flushed the respirometer with fresh water for 15 min and gradually increased current speed to the next trial velocity. A thermostatically controlled cooling system maintained a constant water temperature ($\pm 0.1^\circ\text{C}$) during trials. On occasion, during winter experiments, the cooling system was unable to maintain a constant temperature at high velocities. This occurred at velocities greater than 45 cm s^{-1} and affected experiments only on longnose dace (see Results). We determined the effect of temperature change on oxygen levels by running blank trials. Oxygen consumption by the fish was then obtained by subtracting the blank values from the total change in dissolved oxygen.

We used an electromagnetic flowmeter to calibrate water velocity in the swimming chamber. The flowmeter head blocked 15% of the cross-sectional area of the swimming tunnel and may therefore have inhibited flow somewhat. The tapered, teardrop shape of the meter head should have minimized turbulence. The flow sensors were located at the extreme upstream end of the meter head, so water velocity was measured before flow was constricted at the widest part of the meter head. The water velocity experienced by the fishes tested, however, may have been somewhat different from that recorded by the flowmeter because the diameter of the fishes was less than that of the head of the flowmeter. The shape of the head of the flowmeter and the location of the flow sensors would make this difference very small, however. Corrections for the blocking effect of a fish or other body in an enclosed chamber are discussed by Webb (1975).

During experiments, fishes occasionally became pinned against the downstream retaining screen. When this occurred we (1) decreased flow velocity, (2) momentarily reversed flow to push the fish off the screen, and (3) brought the velocity back up to the trial velocity. If the fish again failed to hold position, the procedure was repeated. At the third failure we assumed that the fish was no longer able to hold its position; the experiment was terminated and this velocity was recorded as the maximum trial velocity. The fish was then removed from the respirometer, weighed ($\pm 0.1\text{ g}$), measured (standard length $\pm 1\text{ mm}$), and sexed by visual inspection of the gonads. Individuals were tested only once. We ran a blank respirometry trial at the end of each day to measure microbial oxygen consumption. This change in DO value, typically zero, was subtracted from all trials run that day.

To minimize the effects of anaerobic metabolism during experiments, we allowed each fish approximately 15 min after each velocity increment to

adjust to the new trial velocity before measuring oxygen consumption (see Jones and Randall 1978). Although it is generally accepted that fishes swim aerobically at low to moderate velocities (see Jones and Randall 1978), anaerobiosis may be significant during burst swimming and at velocities near the fish's maximum attainable velocity (Webb 1971; Driedzic and Kiceniuk 1978). Burst swimming was not a significant factor in our experiments because fishes fatigued rapidly when burst swimming began, and we subsequently terminated experiments. However, oxygen consumption at velocities near the maximum attainable velocity may underestimate the actual costs while swimming.

Statistical Analyses

We regressed seasonal oxygen consumption data (dependent variable; $\text{mg O}_2 \text{ g}^{-1} \text{ h}^{-1}$), on relative velocity (independent variable; $L \text{ s}^{-1}$, where L = body length) with either a linear or an exponential curve using least-squares regression (Zar 1984). The model with the lowest error sum of squares was selected as the model of best fit. For longnose dace, however, a split-line model yielded the best fit (see Nickerson, Facey, and Grossman [1989] for a discussion of the application of this technique to physiological data). We compared regression equations using the Student's t -statistic ($\alpha = 0.05$) and adjusted α for multiple comparisons using the Dunn-Sidak procedure (Ury 1976). When possible, we separated results for males and females and compared them statistically. It is very difficult to measure fishes' standard metabolic rates because many fishes do not remain inactive, even in a static respirometer. Therefore, we estimated standard metabolic rates by extrapolating regressions to zero velocity. This method has been widely used in fish respiration studies (see, e.g., Brett 1965; Rao 1968; Farmer and Beamish 1969; Beamish 1970; Fry 1971; Kruger and Brocksen 1978). We compared mean fish L 's and average maximum velocities across seasons, using Tukey-Kramer tests to identify significant pairwise differences ($\alpha = 0.05$).

Results

There were no significant seasonal differences in L or wet weight for any species tested (table 1).

TABLE 1
Standard lengths and wet weights of fishes tested

	Spring		Summer		Fall		Winter	
	Standard Length (mm)	Wet Wt (g)	Standard Length (mm)	Wet Wt (g)	Standard Length (mm)	Wet Wt (g)	Standard Length (mm)	Wet Wt (g)
Rainbow trout	97 (10)	15.7 (6.2)	78 (31)	9.9 (10.9)	78 (6)	8.3 (2.2)	94 (8)	13.0 (3.4)
Rosyside dace	63 (3)	4.9 (1.0)	67 (5)	5.4 (1.2)	63 (5)	4.4 (1.2)	67 (6)	5.3 (1.8)
Mottled sculpin . .	66 (6)	8.1 (3.0)	66 (9)	7.3 (3.5)	71 (8)	9.7 (1.0)	63 (8)	6.7 (.8)
Longnose dace . .	71 (4)	5.9 (.2)	70 (8)	5.4 (.6)	78 (13)	7.7 (1.3)	78 (10)	7.6 (.9)

Note. Values are means, with 1 SD below in parentheses. There were no significant differences in standard lengths or weights among seasons (Tukey-Kramer test, $\alpha = 0.05$) for any species.

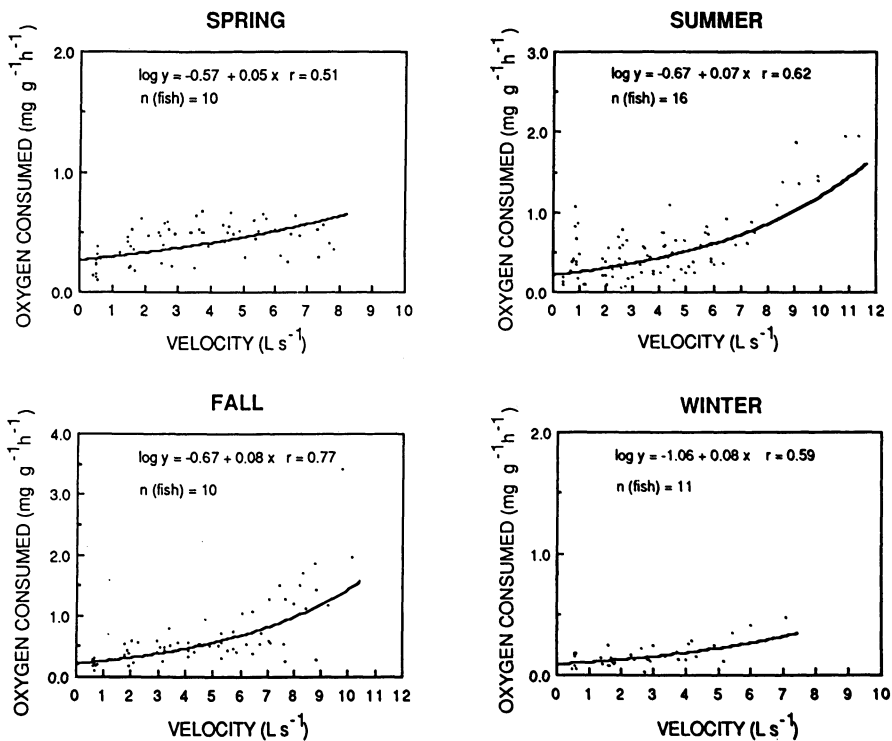


Fig. 1. Seasonal regressions of oxygen consumption (y) and relative velocity (x) for rainbow trout. Relative velocities (L s^{-1}) may be converted to absolute velocities (cm s^{-1}) for the “average” fish by using the mean standard lengths given in table 1. Results for hypothesis tests on slopes are as follows: spring, $P < 0.001$; summer, $P < 0.001$; fall, $P < 0.001$; winter, $P < 0.001$.

Rainbow Trout

Rainbow trout held position by swimming, except for several fish during winter and spring that were able to hold position without swimming at 5 and 15 cm s^{-1} . There was no significant difference, however, in the mean oxygen consumption between swimmers and nonswimmers at either velocity during either season ($P > 0.15$ in all comparisons). Therefore, all data are included in all analyses.

Rainbow trout oxygen consumption increased exponentially with increasing velocity during all seasons (fig. 1). Although fish were tested at identical temperatures and photoperiods during spring and fall, the slope of the fall regression was significantly greater than that of the spring regression ($P < 0.05$). Standard metabolic rate of rainbow trout was significantly lower in winter than during the other seasons ($P < 0.05$). Visual inspection of

TABLE 2
Maximum test velocity ($L\ s^{-1}$) of fishes tested

	Spring	Summer	Fall	Winter
Rainbow trout	6.0 ^{a,b} (2.0)	6.7 ^b (2.5)	8.3 ^b (1.4)	4.1 ^a (1.4)
Rosyside dace	7.5 ^a (2.0)	8.1 ^a (2.4)	6.7 ^a (2.0)	3.4 ^b (1.7)
Mottled sculpin . . .	4.2 ^a (1.4)	5.8 ^b (1.2)	5.1 ^{a,b} (1.3)	4.5 ^a (1.2)
Longnose dace . . .	10.3 ^a (.9)	9.7 ^a (2.3)	9.7 ^a (1.5)	8.5 ^a (1.6)

Note. Values are means, with 1 SD below in parentheses. For each species, seasons with the same superscript letter were not significantly different from one another (Tukey-Kramer test, $\alpha = 0.05$). Sample sizes are presented in figs. 2–6.

gonads showed no evidence of sexual maturity during any season. The mean maximum velocity at which trout were able to hold position (table 2) was highest during fall ($8.3\ L\ s^{-1}$) and lowest in winter ($4.1\ L\ s^{-1}$).

Rosyside Dace

All rosyside dace held position by swimming. The relationship between oxygen consumption and velocity in rosyside dace varied widely among seasons (fig. 2). During spring and winter, regression slopes were not significantly different from zero. Oxygen consumption increased exponentially with velocity in summer, whereas a linear model provided the best fit for fall data. Although temperature and photoperiod were the same during spring and fall, the slope of the fall regression was significantly greater than the spring regression slope ($P < 0.05$). Standard metabolic rates were significantly higher in the fall than in the other seasons, and higher in spring than in summer ($P < 0.05$). Most individuals showed signs of gonadal maturation in the fall, winter, and spring. During the fall, males had higher metabolic rates than females (fig. 3), whereas during the other seasons males and females were not significantly different ($P > 0.05$). Rosyside dace could not swim as fast during winter as they could during spring, summer, and fall (table 2); during the fall, males could not swim as fast as females (fig. 3).

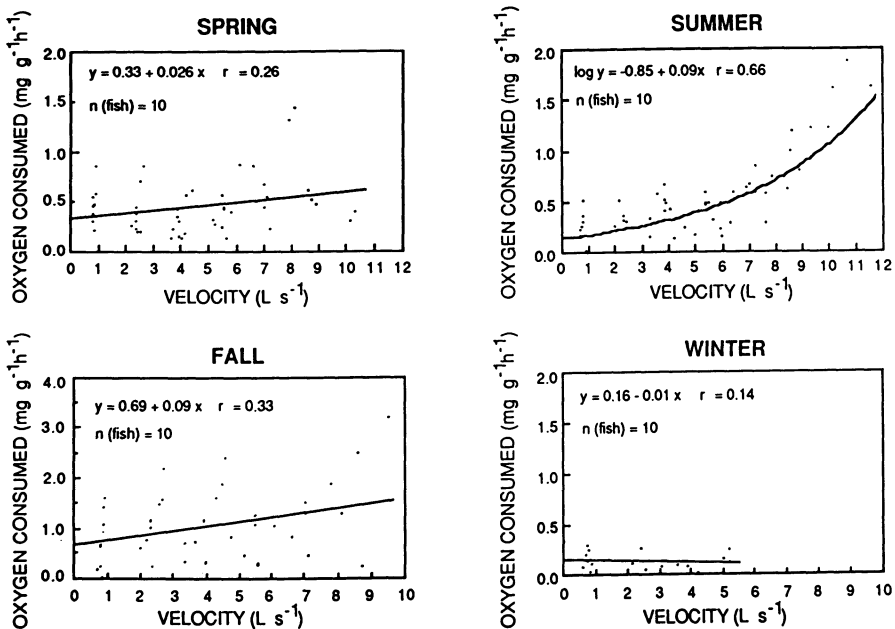


Fig. 2. Seasonal regressions of oxygen consumption (y) and relative velocity (x) for rosyside dace. Results for hypothesis tests on slopes are as follows: spring, $P > 0.05$; summer, $P < 0.001$; fall, $P < 0.05$; winter, $P > 0.05$. For further information see fig. 1.

Mottled Sculpin

Mottled sculpin held position without swimming. When the current was strong enough to sweep them to the rear of the swimming chamber, they attempted several bursts of swimming to regain position but then were pinned against the downstream retaining screen. Mottled sculpin showed little change in oxygen consumption with velocity (fig. 4). Regression slopes were positive in summer ($P < 0.05$), zero in the fall, and slightly negative in spring ($P < 0.05$) and winter ($P < 0.05$). A comparison of slopes indicated that the summer slope was significantly different ($P < 0.05$) from either spring or winter slopes. Slopes from the latter two seasons, however, were not significantly different. Standard metabolic rates were significantly higher in spring than in summer, fall, and winter ($P < 0.05$); the fall value also was significantly higher than the winter value ($P < 0.05$). Mottled sculpin could not hold position at velocities above $6\text{--}8\text{ L s}^{-1}$ during any season, and mean seasonal maximum velocities did not differ significantly (table 2).

We observed sex-related differences in oxygen consumption–relative velocity regressions during summer; the slope for males was positive ($P < 0.05$), whereas the slope for females was not significantly different from

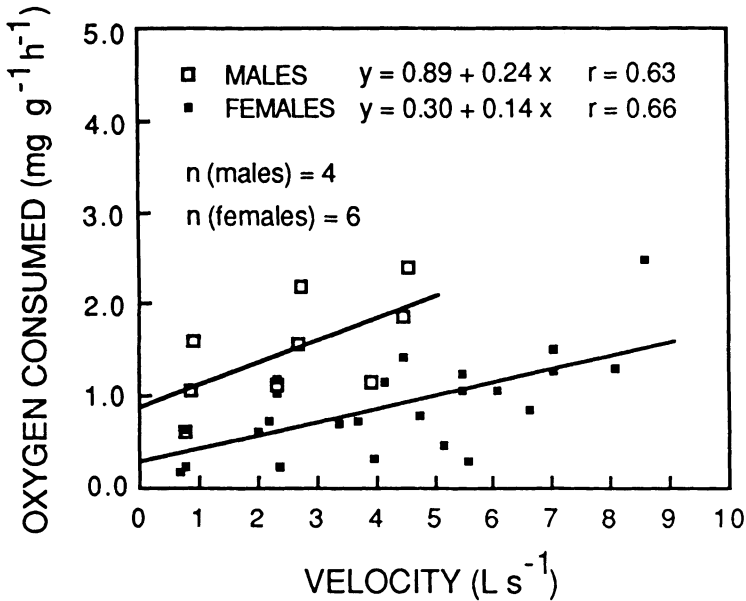


Fig. 3. Regressions of oxygen consumption (y) and relative velocity (x) for male and female rosyside dace during the fall. Results for hypothesis tests on slopes are as follows: males, $P > 0.05$; females, $P < 0.001$. For further information see fig. 1.

zero (fig. 5). There were no significant differences between sexes during the remaining seasons. Mottled sculpin showed gonadal recrudescence during all seasons except summer, but gonads exhibited the greatest development in spring.

Longnose Dace

Longnose dace held position without swimming at low to moderate velocities and swam at high velocities. Oxygen consumption data for spring, summer, and fall were best described by split-line models (fig. 6). During these seasons, oxygen consumption did not change significantly until velocities reached approximately 8 L s^{-1} . At higher velocities, oxygen consumption increased sharply (fig. 6). In winter, oxygen consumption did not vary with velocity, perhaps because only one of the fish tested used a measurable amount of oxygen at velocities greater than 35 cm s^{-1} . The remaining individuals, however, held position at velocities up to at least 55 cm s^{-1} . The increase in temperature at velocities above 45 cm s^{-1} during the winter trials did not measurably increase the oxygen consumption of the longnose dace tested. We did not detect any significant seasonal differences in either stan-

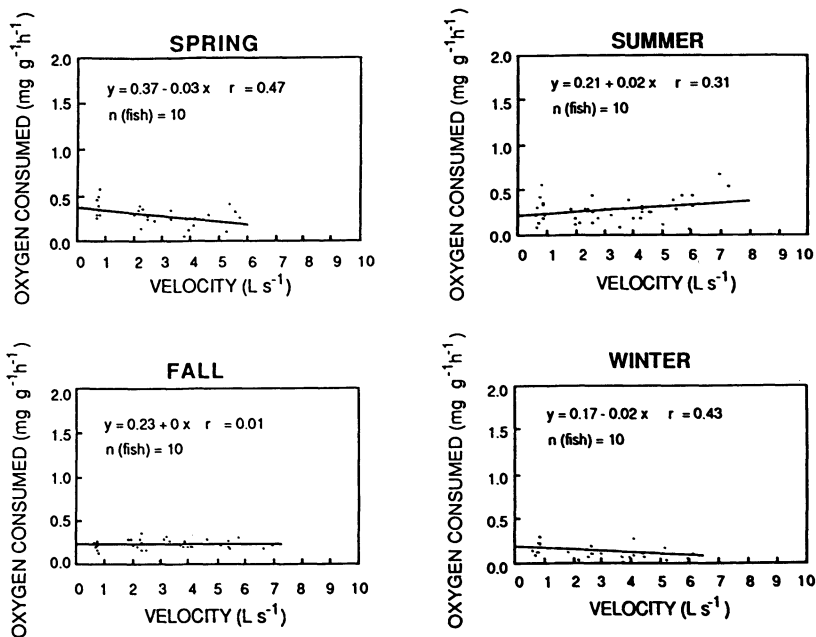


Fig. 4. Seasonal regressions of oxygen consumption (y) and relative velocity (x) for mottled sculpin. Results for hypothesis tests on slopes are as follows: spring, $P < 0.01$; summer, $P < 0.05$; fall, $P > 0.05$; winter, $P < 0.05$. For further information see fig. 1.

standard metabolic rate or mean maximum test velocities for longnose dace ($P > 0.05$; table 2).

Sex-related differences in oxygen consumption rates only were detected during spring (fig. 7). The join point (i.e., the intersection of the split-line regression; see fig. 6) for males corresponded to approximately 7.3 L s^{-1} , whereas the join point for females corresponded to approximately 8.6 L s^{-1} . Neither regression slopes nor standard metabolic rates were significantly different for males and females below their respective join points ($P > 0.05$). Above the join point, however, females exhibited a much greater increase in oxygen consumption with increasing velocity than did males.

Discussion

Seasonal Changes in Metabolic Rates

Many studies have shown that oxygen consumption rates of fishes increase with increasing acclimation temperature (see, e.g., Duthie 1982; Vondracek

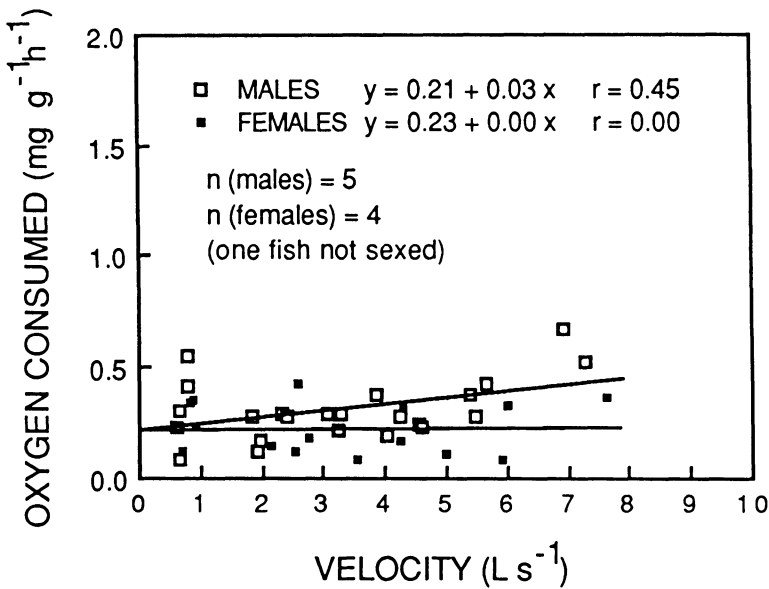


Fig. 5. Regressions of oxygen consumption (y) and relative velocity (x) for male and female mottled sculpin during summer. Results for tests on slopes are as follows: males, $P < 0.05$; females, $P > 0.05$. For further information see fig. 1.

et al. 1982; Cech et al. 1985; Eccles 1985; Lemons and Crawshaw 1985; Du-Preez et al. 1986; McClanahan et al. 1986). However, fishes often were acclimated to different temperatures during a single season under otherwise identical laboratory conditions. The observed temperature effects, therefore, may not represent true seasonal changes in metabolic costs. Metabolic cost values for this study, however, were collected from fishes acclimatized in the natural environment, and therefore should accurately measure seasonal metabolic rates.

During our study, all species exhibited seasonal changes in the relationship between oxygen consumption and velocity. If temperature were the primary factor regulating metabolic rate, then we might expect oxygen consumption to have been lowest in winter (5°C), highest in summer (15°C), and moderate in spring and fall (10°C); spring and fall regressions would not have been significantly different. This was not the case (figs. 1, 2, 4, 6). Standard metabolic rates were higher in spring than in summer for rosyside dace and mottled sculpin. In addition, spring and fall regressions were significantly different for all species ($P < 0.05$). Consequently, our results suggest that fishes may show seasonal changes in metabolic rate independent of temperature. Similar conclusions have been drawn from studies of rain-

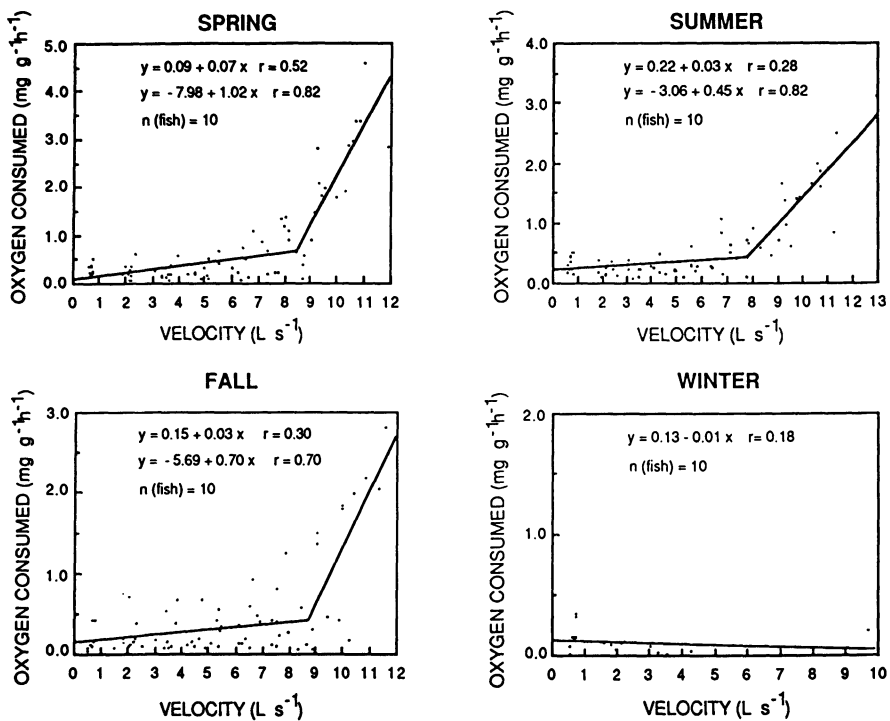


Fig. 6. Seasonal regression of oxygen consumption (y) and relative velocity (x) for longnose dace. The join points (the point at which the two lines meet) for seasonal split-line regressions correspond to the following relative velocities: spring, 8.4 L s^{-1} ; summer, 7.8 L s^{-1} ; fall, 8.7 L s^{-1} . Results for hypothesis tests on slopes are as follows: winter, $P > 0.05$; other seasons all had nonsignificant ($P > 0.05$) slopes below their join points, and significant ($P < 0.001$) slopes above their join points. For further information see fig. 1.

bow trout (Dickson and Kramer 1971) and pumpkinseed sunfish (*Lepomis gibbosus*; Evans 1984).

Although most species displayed a positive relationship between oxygen consumption and velocity, mottled sculpin data yielded significantly negative slopes during winter and spring ($P < 0.05$; fig. 4). This may have been caused by substantial interindividual variation in behavior that affected oxygen consumption observed at the lowest test velocity (5 cm s^{-1}). At 5 cm s^{-1} some mottled sculpin faced upstream whereas others oriented downstream or perpendicular to the current. This difference may have affected the amount of energy required to draw water into the mouth. At higher velocities, however, fish consistently faced upstream. If we omitted the data at 5 cm s^{-1} from all seasons, (1) the slopes during spring, fall, and winter would not be significantly different from zero, (2) the slope during summer would

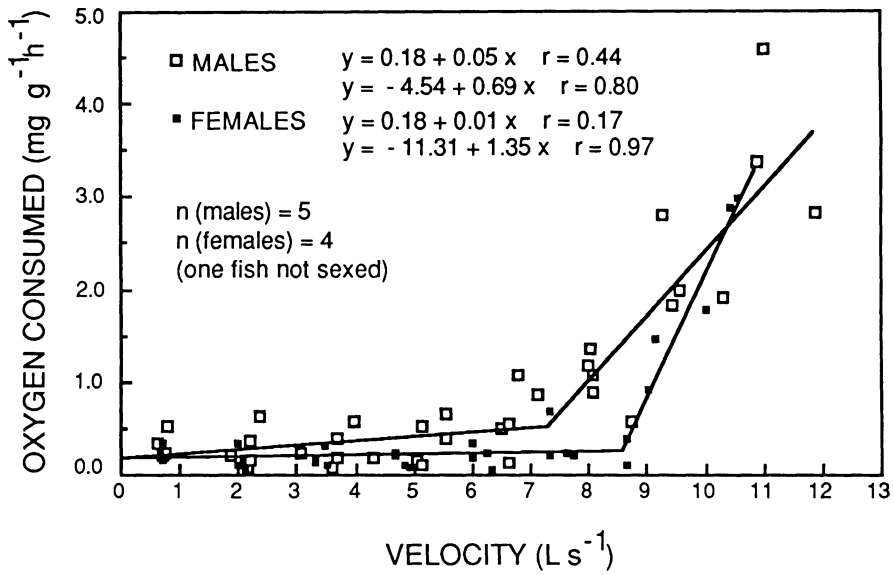


Fig. 7. Regressions of oxygen consumption (y) and relative velocity (x) for male and female longnose dace during spring. The join points correspond to the following relative velocities: males, 7.3 L s^{-1} ; females, 8.6 L s^{-1} . Results for hypothesis tests on slopes are as follows: both sexes showed nonsignificant ($P > 0.05$) slopes below their join points, and significant ($P < 0.001$) slopes above their join points. For further information see fig. 1.

be positive (0.04), and (3) the standard metabolic rates during spring ($0.287 \text{ mg O}_2 \text{ g}^{-1} \text{ h}^{-1}$) and fall (0.249) would be significantly greater ($P < 0.05$) than the standard metabolic rates during summer (0.107) and winter (0.116).

The higher spring standard metabolic rates of rosyside dace and mottled sculpin may have been caused by physiological preparation for spawning, which occurs during spring and summer. Gonadal recrudescence in rosyside dace began in the fall, when standard metabolic rates were highest; standard metabolic rate decreased in winter, and increased again in spring. A similar pattern has been noted for pumpkinseed sunfish in northern temperate ponds and streams (Roberts 1964). The large scatter in the rosyside dace data for spring and fall may have been due, in part, to interindividual differences in the degree of reproductive maturity. Other fishes also exhibit elevated metabolic rates prior to and during the spawning season (Beamish 1964; Roberts 1964; Dickson and Kramer 1971; Burns 1975). Although longnose dace also spawn in spring and summer, we did not detect significant interseasonal differences in standard metabolic rates. None of the rainbow trout exhibited gonadal recrudescence; hence, reproductive condition probably did not affect interseasonal differences in metabolic rate.

Our estimates of standard metabolic rates for rainbow trout were similar to those reported in other studies. In spring (10°C), summer (15°C), fall (10°C), and winter (5°C) trout used an estimated 0.27, 0.22, 0.22, and 0.07 mg O₂ g⁻¹ h⁻¹, respectively. The equations reported by Rao (1968) yielded values of 0.186 and 0.089 mg O₂ g⁻¹ h⁻¹ at 15° and 5°C for trout of the size we studied. Larger rainbow trout (80–400 g) studied by Dickson and Kramer (1971) exhibited standard metabolic rates of approximately 0.04 (5°C), 0.05 (10°C), and 0.07 (15°C) mg O₂ g⁻¹ h⁻¹; these may have been lower because the fish studied by Dickson and Kramer were larger than those that we used. Webb (1971) also studied larger rainbow trout (approximately 250–300 g) and estimated standard metabolic rate to be 0.07 mg O₂ g⁻¹ h⁻¹ at 15°C.

Variability in our data occasionally made interpretation difficult. Such variability, however, is not atypical in studies of fish metabolic rates (Fry 1971). In addition, we believe this to be a real phenomenon, not an experimental artifact. We used fishes taken from wild populations. Therefore we could not control genetic background or rearing conditions. Data derived from hatchery- or laboratory-reared specimens might have decreased variability, but probably would not have been representative of the metabolic cost of holding position among wild fishes.

Another factor complicating interpretation of our data was that for each species the maximum velocity at which fish could hold position varied among individuals. The paucity of data values at the highest velocities resulted in regressions that underestimate the actual oxygen consumption of fishes that were able to hold position at these velocities. This was especially evident for rainbow trout in summer, fall, and winter (fig. 1); rosyside dace in summer (fig. 2); and mottled sculpin in summer (fig. 4).

Seasonal Changes in Swimming Ability

Rainbow trout and rosyside dace, both water-column species, showed greatly decreased swimming ability in winter (table 2; figs. 1, 2). This response also has been observed in other species subjected to low temperatures (Brett, Hollands, and Alderdice 1958; Griffiths and Alderdice 1972; Bernatchez and Dodson 1985), perhaps because of a decrease in the speed of muscle contraction (Brill and Dizon 1979; Sidell and Johnston 1985; Moerland and Sidell 1986).

Intersexual Differences in Metabolism and Swimming Ability

Our results also indicate that males and females of the same species may differ in metabolic rates and swimming ability during some seasons but not

others. For example, male rosyside dace consumed more oxygen than females during the fall, but this relationship was not observed in any other season (fig. 3). During the fall, males could not swim as fast as females, nor could they swim as well as males in other seasons. Perhaps the additional demand of gonad development is greater among males during the fall, thereby increasing metabolic costs and decreasing aerobic scope. Male mottled sculpin showed a slight but significant increase in metabolic costs as velocity increased during summer, whereas females had nonsignificant regression slopes (fig. 5). Regressions of oxygen consumption versus velocity for male and female longnose dace were significantly different only in spring. The relative velocity of the estimated join point for males (7.3 L s^{-1}) was lower than that for females (8.6 L s^{-1}), suggesting that males may switch to active swimming at a lower relative velocity during this season. At velocities above the join point, however, females exhibited a much greater increase in metabolic cost than did males. To our knowledge, this is one of the first studies documenting intersexual differences in metabolic rates of cyprinids. Sex-related differences in metabolic rate have been noted in brown trout (*Salmo trutta*) (Beamish 1964) and rainbow trout (Dickson and Kramer 1971). Although our data do not provide an explanation for the sex-related differences observed in our study, this may be an important area for future research. Intersexual differences in metabolic rates could be a source of error in energy-budget predictions and cost-benefit models that do not differentiate between sexes.

Adaptation of Benthic Fishes

Stream-dwelling benthic fishes exhibit a variety of adaptations that allow them to remain stationary without actively swimming in high-velocity microhabitats. The split-line models of oxygen consumption by longnose dace indicated that there were no significant changes in the metabolic cost required to hold position at velocities between 0 and approximately 8 L s^{-1} . At higher velocities, however, the cost of holding position increased rapidly. Longnose dace are benthic fishes that use their fusiform, somewhat depressed shape and large, extended pectoral fins to hold themselves against the substrate. Even on the smooth bottom of the respirometer, longnose dace could hold position over a range of velocities without actively swimming. The extended pectoral fins may help to force the fish toward the bottom in fast water, as has been reported for young Atlantic salmon (*Salmo salar*) (Keenleyside 1962), stream-dwelling suckers (family Catostomidae) (Lundberg and Marsh 1976), and the common bully (*Gobiomorphus cotidianus*) (Davison 1983). As water velocity increased, longnose dace exhibited

a change in posture similar to that reported for the fantail darter (*Etheostoma flabellare*) (Matthews 1985). These species tended to incline forward so that the tail was elevated and the ventral surface was off the bottom.

Unlike longnose dace, mottled sculpin are not good swimmers. They do, however, hold their pectoral fins out laterally when in a current and use their large pectoral fins to cling to the substrate. Mottled sculpin held position without swimming, and therefore showed little or no change in metabolic cost at velocities up to $6\text{--}8\text{ L s}^{-1}$. They were unable to hold position at higher velocities. Their ability to hold position is similar to that of the common bully of New Zealand which can hold position without swimming at water velocities of up to 5 L s^{-1} ; efforts to swim at higher velocities resulted in rapid fatigue (Davison 1983).

Split-line regressions provided the best fit for oxygen consumption data for longnose dace (figs. 6, 7). To our knowledge, however, split-line relationships between oxygen consumption and velocity have not been reported for other fishes. Similar patterns may exist in other benthic stream fishes that can passively hold position up to a critical current velocity. Further study of benthic stream fishes is needed to determine whether the relationship between oxygen consumption and velocity for this group of fishes is best described by split-line models.

Our results indicate that seasonal changes in metabolic rates may be affected by factors other than temperature (e.g., reproductive condition)—a finding consistent with several other studies (Wells 1935; Roberts 1964; Dickson and Kramer 1971; Burns 1975; Evans 1984). It may be necessary for fishes to be subjected to natural changes in temperature, photoperiod, and other external stimuli in order to induce seasonal physiological changes such as gonadal recrudescence (Fry 1971; Somero and Hochachka 1971). Thus, estimates of seasonal metabolic costs derived by subjecting specimens collected during a single season to different temperature regimes may be inaccurate. A lack of appropriate data, however, has caused some investigators to apply metabolic rates from temperature-acclimation studies to energetic and ecological models (e.g., Feldmeth and Jenkins 1973; Fausch 1984). Models that do not account for seasonal physiological changes are likely to be misleading. We suggest that researchers could increase the accuracy of seasonal metabolic cost estimates through the use of naturally acclimated individuals.

Acknowledgments

We thank the following for their assistance and support, for reviewing the manuscripts, and for manuscript preparation: J. Barrett, F. W. H. Beamish, J.

Cech, M. Crawford, J. Durniak, C. Faanes, M. Flood, S. Floyd, M. Freeman, B. Goldowitz, S. Hales, B. Harvey, G. S. Helfman, J. Hill, J. Kerr, H. Li, W. Matthews, B. Mullen, A. Nelms, D. Stouder, and L. Willoughby. In addition, we acknowledge the invaluable support of the staff of the U.S. Forest Service Coweeta Hydrologic Laboratory. This research was supported by the School of Forest Resources at the University of Georgia and the U.S. Department of Agriculture's McIntire Stennis Program (grant GEO-0035-MS to G.D.G.).

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